Test 4 Practice — Magnetism & Induction

These are modeled on the hardest problems from Written HW 7, HW 8, and HW 9 – same problem types, new numbers. Every formula you need is on the Magnetism equation sheet; have it open while you work. Each pair of problems below tests the same idea but asks for a different unknown, so skim the whole question before you start plugging in. The answer key is on the last page.

Part A — Magnetic Fields, Forces & Motion

1. A long straight wire carries a current of 150 A. At what distance from the wire does the magnetic field strength equal 3.0 mT?

2. A long straight wire is 5.0 cm from a point where the magnetic field strength is measured to be 80 μT. What current flows in the wire?

3. A 50-turn circular coil of radius 4.0 cm carries a current of 2.0 A. Find (a) the coil's magnetic dipole moment, and (b) the magnetic field strength at the center of the coil.

4. A single-turn circular loop of radius 3.0 cm carries current I. The magnetic field at the loop's center is measured to be 42 μT. What current flows in the loop?

5. An electron travels at 2.0×10^6 m/s between two parallel plates separated by 2.0 cm and charged by a 400 V battery (top plate positive). What magnetic field strength and direction will let the electron pass between the plates without being deflected?

6. An antiproton (same mass as a proton, $q = -e$) moves at 400 m/s to the left through a region with uniform fields $\vec{E} = 1500$ V/m (pointing up, in the plane of the page) and $\vec{B} = 3.0$ T (into the page). Find the magnitude and direction of the antiproton's acceleration at this instant.

7. A uniform 40 mT magnetic field points in the $+z$ direction. A proton enters the field moving at 2.0×10^6 m/s, at 40° above the xy-plane. Find the radius r and the pitch p of the proton's helical trajectory.

8. A proton moves through a uniform 25 mT magnetic field on a helical path with radius 2.0 cm and pitch 8.0 cm. Find the proton's speed v and the angle above the xy-plane at which it entered the field.

Part B — Electromagnetic Induction

9. A 15-cm-long wire is pulled along a U-shaped conducting rail in a perpendicular magnetic field. The total resistance of the wire and rail is 0.40 Ω. Pulling the wire at a steady speed of 3.0 m/s causes 6.0 W of power to be dissipated in the circuit. Find (a) the pulling force, and (b) the magnetic field strength.

10. A 20-cm-long wire slides along a U-shaped conducting rail in a 1.5 T field perpendicular to the rail's plane, pulled at a constant 5.0 m/s. The total resistance is 0.60 Ω. Find (a) the induced emf and current, (b) the power dissipated, and (c) the pulling force required.

11. A square conducting loop, 15 cm on each side, is positioned so that exactly 60% of its area lies within a 0.25 T magnetic field (directed into the page); the rest of the loop is in a field-free region. (a) What is the magnitude of the magnetic flux through the loop? (b) If the field strength *increases*, what is the direction of the induced current in the loop?

12. A bar magnet's south pole is pushed toward a stationary conducting loop, face-on. As the magnet approaches, is a current induced in the loop? If so, use Lenz's law to determine whether the near face of the loop becomes a north or south magnetic pole.

13. A 500-turn coil of wire 2.0 cm in diameter sits in a magnetic field that increases from 0.050 T to 0.200 T in 25 ms. The coil's axis is parallel to the field. What is the induced emf of the coil?

14. A 50-turn coil, 4.0 cm in diameter, has a resistance of 2.0 Ω. A magnetic field perpendicular to the coil increases at a steady rate of 0.80 T/s. Find (a) the induced emf, and (b) the induced current.

15. The magnetic field inside a circular region of radius 4.0 cm points into the page and is decreasing at the rate 0.25 T/s. (a) What is the magnitude of the induced electric field at a point 2.0 cm from the center? (b) Is the induced electric field clockwise or counterclockwise? (c) A proton released from rest at that point momentarily feels a force. What is the magnitude of its initial acceleration?

16. At a point 5.0 cm from the center of a circular field region, the induced electric field has magnitude 0.030 V/m. (a) At what rate is the magnetic field changing? (b) An electron released from rest at a point 2.0 cm from the center (same field region) feels what initial acceleration?

ANSWER KEY — FLIP THE PAGE OVER (OR STAND ON YOUR HEAD)

Part A — Magnetic Fields, Forces & Motion

1. $r = 1.0$ cm

2. $I = 20$ A

3. (a) $\mu = 0.50\ \text{A} \cdot \text{m}^2$ (b) $B = 1.6$ mT

4. $I = 2.0$ A

5. $B = 10.0$ mT, into the page

6. $F_{\text{net}} = 4.8 \times 10^{-17}$ N, down; $a = 2.9 \times 10^{10}\ \text{m/s}^2$, down

7. $r = 0.40$ m; $p = 2.1$ m

8. $v \approx 5.7 \times 10^4$ m/s; $\theta \approx 33^\circ$

Part B — Electromagnetic Induction

9. (a) $F_{\text{pull}} = 2.0$ N (b) $B \approx 3.4$ T

10. (a) $\mathcal{E} = 1.5$ V, $I = 2.5$ A (b) $P = 3.75$ W (c) $F_{\text{pull}} = 0.75$ N

11. (a) $\Phi \approx 3.4 \times 10^{-3}$ Wb (b) counterclockwise

12. Yes; the near face becomes a south pole (repelling the approaching magnet)

13. $\mathcal{E} \approx 0.94$ V

14. (a) $\mathcal{E} \approx 50$ mV (b) $I = 25$ mA

15. (a) $E = 2.5 \times 10^{-3}$ V/m (b) clockwise (c) $a \approx 2.4 \times 10^5\ \text{m/s}^2$

16. (a) $|dB/dt| = 1.2$ T/s (b) $a \approx 2.1 \times 10^9\ \text{m/s}^2$